

Remote Sensing for Wildfire Mapping: A Comprehensive Review of Advances, Platforms, and Algorithms

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Abstract

The use of remote sensing technologies for mapping forest fires has experienced significant growth in recent decades, driven by advancements in remote sensors, processing platforms, and artificial intelligence algorithms. This study presents a review of 192 scientific articles published between 1990 and 2024, selected using PRISMA criteria from the Scopus database. Trends in the use of active and passive sensors, spectral indices, software, and processing platforms as well as machine learning and deep learning approaches are analyzed. Bibliometric analysis reveals a concentration of publications in Northern Hemisphere countries such as the United States, Spain, and China as well as in Brazil in the Southern Hemisphere, with sustained growth since 2015. Additionally, the publishers, journals, and authors with the highest scientific output are identified. The normalized burn ratio (NBR) and the normalized difference vegetation index (NDVI) were the most frequently used indices in fire mapping, while random forest (RF) and convolutional neural networks (CNN) were prominent among the applied algorithms. Finally, the main technological and methodological limitations as well as emerging opportunities to enhance fire detection, monitoring, and prediction in various regions are discussed. This review provides a foundation for future research in remote sensing applied to fire management.

Keywords: burned area mapping; fire severity mapping; fire management; remote sensing; satellite sensors; wildfires



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1. Introduction

Long before humans, fires occurred primarily due to natural causes in the tropics and subtropics [1,2]. Currently, nearly 50% of the Earth's surface is susceptible to fires, with a one-third probability of experiencing frequent and intense burns [3]. Therefore, it is important to review the impact of fires on landscapes considering human interactions, especially during periods of drought and high temperatures, when fires are on the rise [4]. This increase is associated with climate variability, which plays a significant role in the spread of fires [5,6]. This is reflected in the multiple levels of severity, and it is estimated that in the future the likelihood of occurrence will be greater [7]. These occurrences generally happen during the dry season of the year, where there is a greater prolongation and intensity of fires globally, resulting in considerable atmospheric pollution [8,9].

Monitoring fires before, during, and after their occurrence can contribute to the planning of preventive actions in vulnerable communities and mitigate the impact on ecosystems [10]. Traditionally, fire detection was done through human observation from watchtowers, a method prone to fatigue and error [11]. Therefore, terrestrial sensors serve as an alternative to complement in situ observations, although they have limited coverage and require a large number of sensors to monitor extensive areas [12]. At various monitoring scales, a range of geospatial technologies assist in this task, utilizing tools such as remote sensing [13]. This method involves the acquisition of data from active and passive sensors, incorporating innovations over the years that include cloud-based data acquisition and processing and, more recently, the use of artificial intelligence (AI) [14,15].

However, acquiring and processing data from remote sensors possesses certain spatial and temporal limitations as well as susceptibility to weather conditions and cloud coverage (primarily in passive sensors). These factors limit continuous and detailed monitoring of fire occurrences [16]. Notable platforms with passive sensors for these purposes include NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) and Visible Infrared Imaging Radiometer Suite (VIIRS) as well as the combined use of optical and thermal data from the Landsat satellite [17–19] or Sentinel 2 [20]. At lower levels of observation, multispectral and hyperspectral sensors installed on aircraft, helicopters, and unmanned aerial vehicles (UAV) are also employed for fire detection, offering high spatial resolution and greater operational flexibility, albeit with more limited temporal resolution [21,22]. Additionally, active sensors such as synthetic aperture radar (SAR) and light detection and ranging (LiDAR) have proven highly useful in both the detection of fires and the quantification of affected areas post fire, even in the presence of cloud cover or smoke [23,24].

Data obtained from passive remote sensors allow the calculation of spectral vegetation indices (VI), which contribute to the identification, quantification, and comparison of affected areas before and after a fire [25–27]. Among the most commonly used VI for these purposes are the normalized difference vegetation index (NDVI), the normalized burn ratio (NBR), and the soil-adjusted vegetation index (SAVI), which have shown high sensitivity to changes in vegetation cover caused by fire [28]. These indices facilitate the evaluation of changes in vegetation cover, the severity of damage, and post-fire regeneration processes, forming a crucial basis for developing advanced methodologies for the analysis, modeling, and prediction of these events.

Using data obtained from remote sensors and spectral indices (VI), advanced methodologies have been developed to integrate algorithms and predictive models to enhance the detection, monitoring, and management of wildfires, optimizing resources and mitigating their impacts [29]. Within artificial intelligence, machine learning (ML) and deep learning (DL) algorithms stand out for their capacity to process large volumes of satellite data [30,31]. ML algorithms such as artificial neural networks (ANN), support vector machines (SVM), decision trees (DT), and random forests (RF) have proven effective in predicting the occurrence, spread, and intensity of fires [32–37]. DL has demonstrated significant potential in fire detection by automatically learning complex spatial and temporal features, especially when trained on large datasets [38]. Various algorithms, including ResNet50, AlexNet, GoogleNet, VGG16, and MobileNetV2 as well as convolutional neural networks (CNN), recurrent neural networks (RNN), and long short-term memory (LSTM) networks, have been implemented to classify and delineate burnt areas in different regions worldwide [39–43].

Given the diversity of remote sensors, VI, and algorithms applied to forest fire monitoring, it is essential to systematically evaluate how these tools have been used over time and what gaps persist in their implementation. Although satellite imagery of the Earth's surface has been collected since the 1970s [44], global data coverage remained limited until the early 1990s, when standardized and georeferenced datasets such as the National Land

Cover Dataset (NLCD) in the United States became available [45], followed by similar efforts in other countries such as Canada [46], making the 1990s a temporal baseline for this review. This study presents a comprehensive review of research from 1990 to 2024, examining not only the use of sensors and indices but also the geographical distribution of studies, the publishers with the highest number of publications, and the most-cited authors in this field. Additionally, the main challenges are identified, and future directions are proposed to advance the study and management of forest fires through the use of remote sensing technologies.

2. Materials and Methods

In the Scopus bibliographic and citation database (<https://www.scopus.com/>, accessed on 30 December 2024), a comprehensive set of queries was utilized for this review, including terms such as “forecasting” and “modeling”, which, although not strictly limited to wildfire mapping, were included considering that they are often used in studies that involve fire risk assessment, fire spread prediction, or related remote sensing applications. These search strings included terms and expressions such as TITLE-ABS-KEY ((“Forest fire detection” OR “Forest fire” OR “Wildfire” OR “Vegetation fire” OR “Bushfire” OR “Burned area” OR “Fire regime” OR “Burned severity” OR “mountain fire”) AND (“Prediction” OR “Detection” OR “Segmentation” OR “Forecasting” OR “Modeling” OR “Map” OR “Classification” OR “Identification”) AND (“Deep learning” OR “Machine learning” OR “Regression” OR “Natural networks” OR “Artificial intelligence” OR “burn index”)) AND (“Remote sensing” OR “Satellite imaging” OR “Remote observation”) AND PUBYEAR > 1990 AND PUBYEAR < 2024 AND (LIMIT-TO (SRCTYPE, “j”)) AND (LIMIT-TO (PUBSTAGE, “final”)) AND (LIMIT-TO (DOCTYPE, “ar”)) AND (LIMIT-TO (LANGUAGE, “English”)).

Logical operators such as “AND” were utilized between groups of words in titles and topics, whereas “OR” was used between categories within each preceding group, as shown in Figure 1. For example, the “Title” column filters articles containing terms like “Forest fire detection” or “Forest fire” exclusively within the title. Meanwhile, the “Topic” columns expanded the search to include these terms in titles, abstracts, and keywords of the articles. Additionally, specific filtering criteria include the publication range (1990–2024), document type, language, and publication stage.

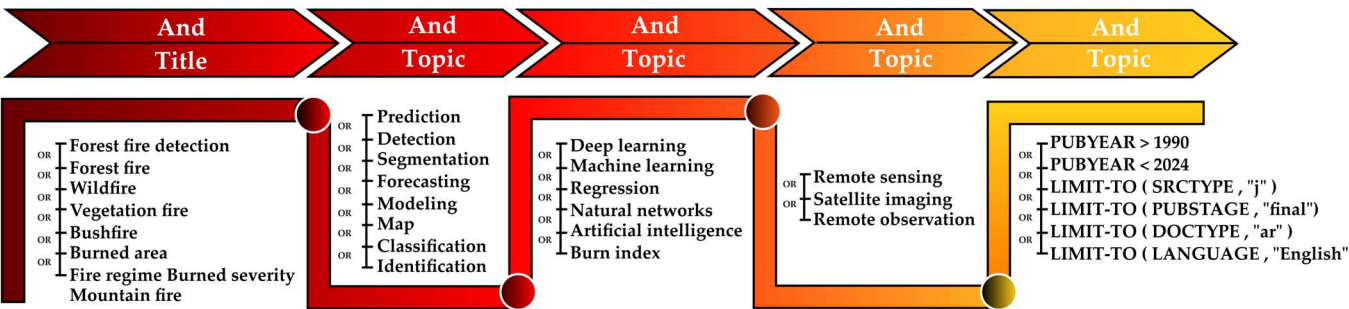


Figure 1. Search query chains include terms, expressions, and logical operators for this review. The key terms in the first column (left) were used in the search titles of the articles, while the words in the subsequent columns were considered to be included in addition to the title, abstract, and keywords of the articles.

The workflow was based on the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) and is visualized in Figure 2, generating a standardized and transparent approach for this review, similar to what has been used in previous studies [47]. Initially, a total of 1608 journal articles were obtained from the Scopus database, with six ar-

ticles excluded for not being methodological studies. Additionally, one article was excluded due to inaccessibility to the full text, preventing the collection of all necessary information for inclusion in the review. The remaining 1601 studies were thoroughly reviewed, considering their relevance to the central theme of forest fire mapping. Articles whose focus did not directly relate to the objective of this review, such as studies exclusively focused on prediction models, severity assessments, fire management, or associated socioeconomic factors, were excluded. The final result was the inclusion of 192 published articles in this comprehensive review.

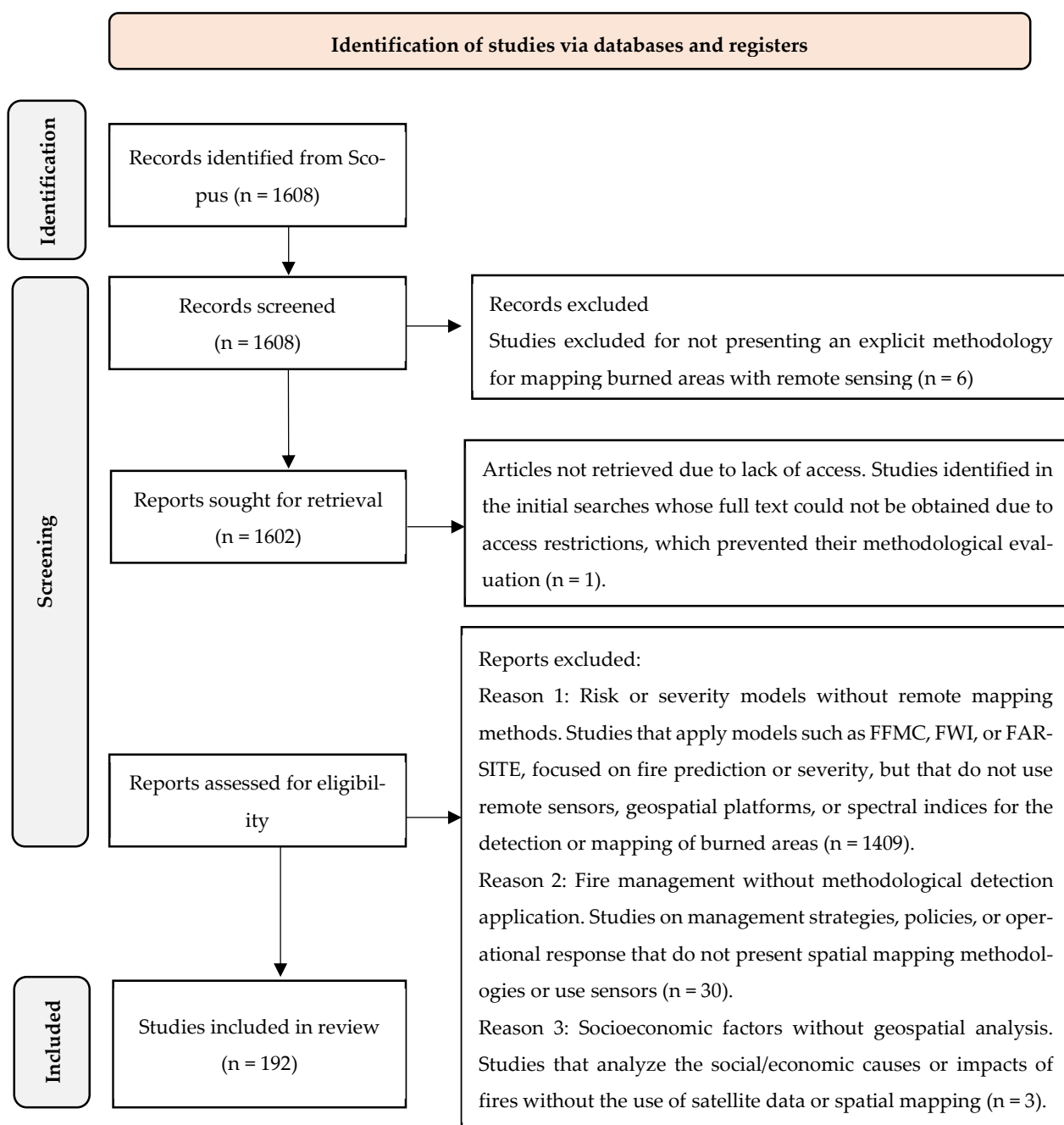


Figure 2. PRISMA-based flow diagram for the selection of relevant articles from advanced search queries in Scopus.

Subsequently, using bibliometric tools such as Bibliometrix (<https://www.bibliometrix.org/home/>), accessed on 12 March 2025) and VOSviewer version 1.6.20 (<https://www.vosviewer.com/>).

vosviewer.com/, accessed on 12 March 2025), the keywords from the final 192 articles were identified. The frequency of these keywords is presented in Figure 3.

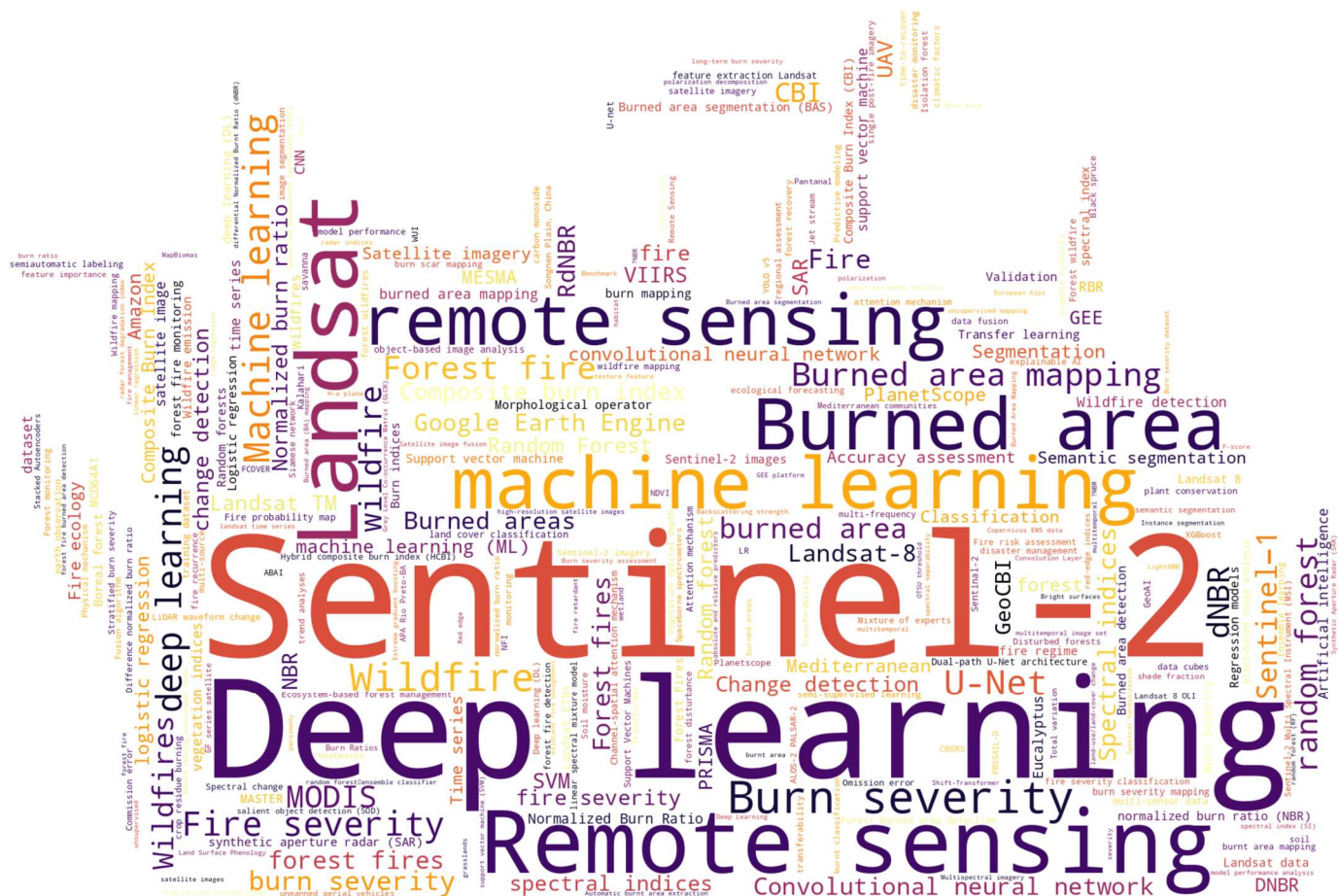


Figure 3. Word cloud of keywords in scientific publications on remote sensing technologies applied to fire mapping (1990–2024). The size of each term represents its frequency of occurrence.

Finally, a database was developed containing 13 attributes as shown in Table 1, consolidating information for each article, including the title, general author data, year, citation, and publication. Additionally, further terms were considered to facilitate the systematic analysis of trends and patterns based on the methodology, sensor, use of drones, satellites, vegetation indices, and platforms used in each study.

Table 1. Variables collected from the studies included in the database for the analysis.

No.	Attribute	Description
1	Authors	Author names
2	Title	Article title
3	Journal title	Title of the journal
4	Year	Year of publication
5	Citations	Number of article citations
6	Affiliation	Author affiliation country
7	Method used	ML, DL, regression, etc.
8	Use of UAVs	Fixed-wing, multirotor, etc.
9	Sensor type	Optical, SAR, etc.

Table 1. Cont.

No.	Attribute	Description
10	VIs used	NDVI, EVI, etc.
11	Platform	GEE, MODIS, etc.
12	Software	ArcGIS, QGIS, etc.
13	Satellites	Landsat, Sentinel, Himawari, etc.

Specifically, the consolidated articles in the database were examined, considering key parameters related to the use of remote sensing for forest fire monitoring. These parameters included the type of spatial platform, the type of sensor, the methodology employed for data analysis, and the software or computational environment used.

3. Results

3.1. Bibliometric Analysis

3.1.1. Number of Publications and Geographic Distribution

A significant 67.2% of the total publications (129 out of 192) since 1997 were concentrated in the last five years (2020–2024), with 2024 having the highest number of records, as depicted in Figure 4. This trend is supported by a positive temporal increase in the number of studies published annually, with a visible and statistically significant rise over time (logarithmic model: $R^2 = 0.72$, $p < 0.001$).

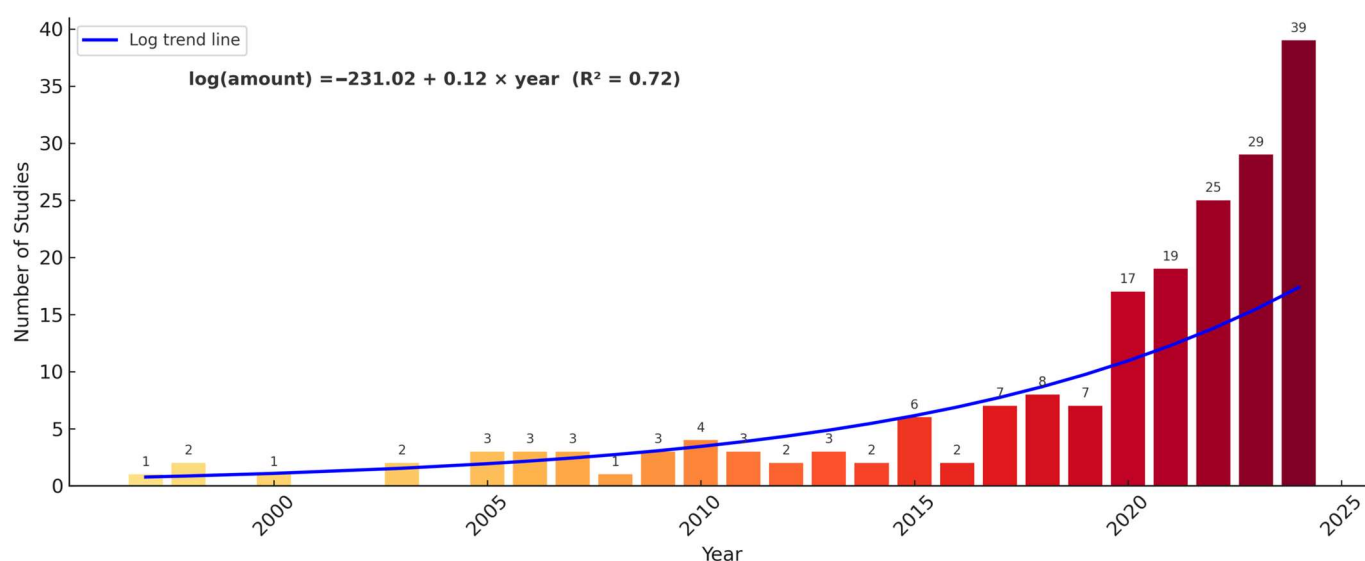


Figure 4. Temporal trend in the number of studies on forest fire monitoring using remote sensing published per year (1997–2024), based on 192 records. A logarithmic regression model was fitted, showing a significant increase over time.

The geographic distribution of the number of studies is shown in Figure 5. Publications from 27 countries were identified regarding technologies and methodologies of remote sensing applied to fire mapping. The country with the highest number of studies (n) was the United States ($n = 41$), followed by Spain ($n = 24$), China ($n = 21$), Brazil ($n = 12$), Portugal ($n = 11$), and South Korea ($n = 10$). Notable mentions include Iran ($n = 8$), Germany, Italy, and Canada ($n = 7$ each) as well as Greece and Australia ($n = 6$ each). Other countries with lesser representation include Sweden ($n = 5$); Turkey and Indonesia ($n = 4$ each); India, Russia, and Chile ($n = 3$ each); France ($n = 2$); and Malaysia, South Africa, Mexico, Zimbabwe, Peru, Belgium, Switzerland, and Israel ($n = 1$ each).

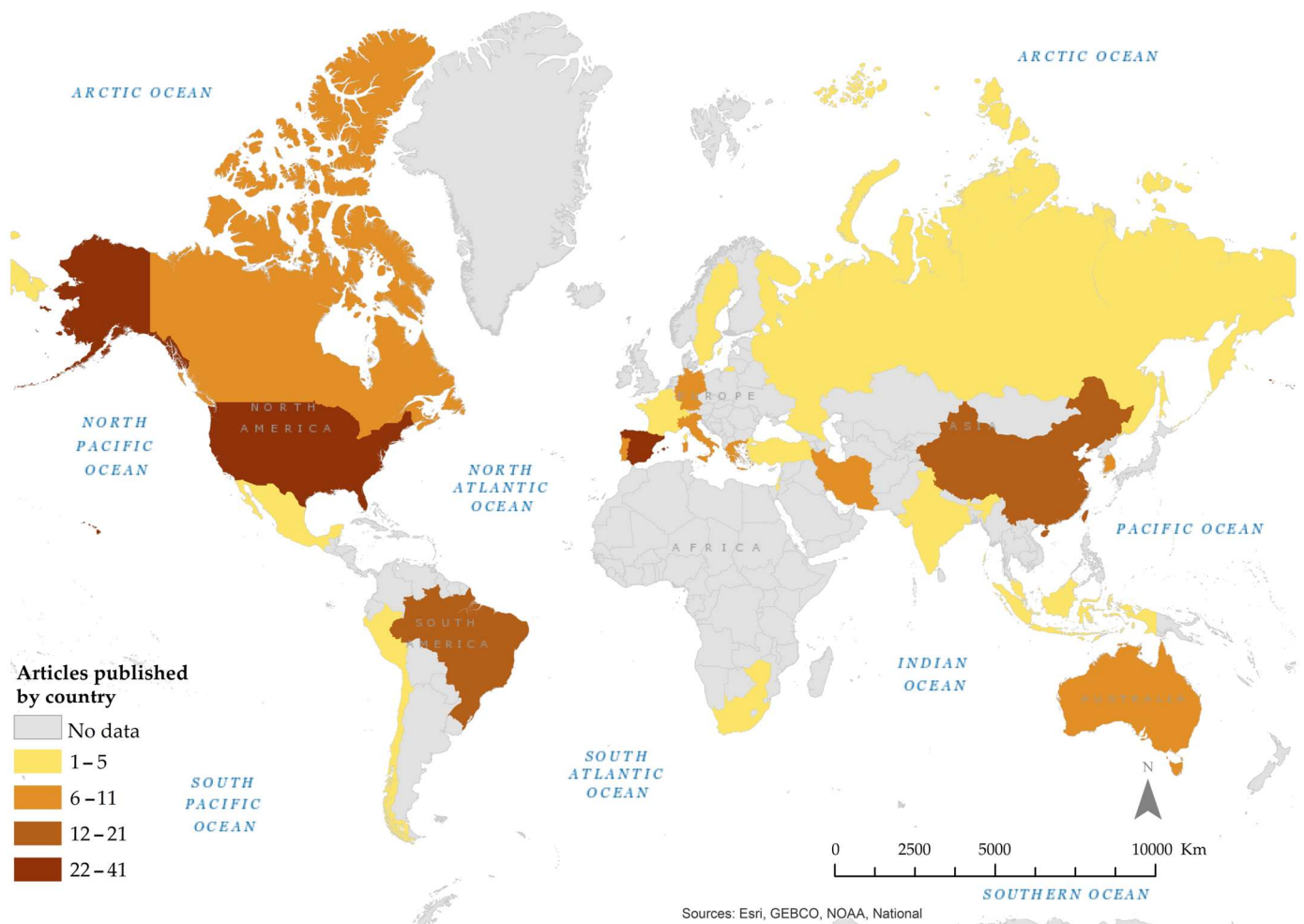


Figure 5. Number of articles on remote sensing applied to fire mapping by country, classified by frequency intervals.

3.1.2. Publishers and Journals

Among the publishers with the highest number of published articles are Multidisciplinary Digital Publishing Institute (MDPI) with 59 articles, Elsevier with 56, Taylor & Francis with 21, and the Institute of Electrical and Electronics Engineers (IEEE) with 10. The Commonwealth Scientific and Industrial Research Organisation (CSIRO) has 6 publications, and 40 articles are distributed among other publishers.

In detail, Figure 6a shows that the journal *Remote Sensing*, published by MDPI, has the highest number of publications ($n = 37$), representing 62.71% of the total published by this publisher. Similarly, *Remote Sensing of Environment* (RSE) stands out with 29 articles (51.79%) published under Elsevier (Figure 6b). *The International Journal of Remote Sensing* (IJRS), published by Taylor & Francis, has 13 publications, representing 61.90% (Figure 6c). The IEEE *Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (JSTAEORS) has published six articles, representing 60% of the total published by IEEE (Figure 6d). Lastly, a diverse group of journals from other publishers covers a total of 40 articles, including *Sensors and Materials*, *Earth's Future*, *Frontiers in Forests and Global Change*, *Photogrammetric Engineering and Remote Sensing*, and *Environmental Science and Pollution Research International*, among others.

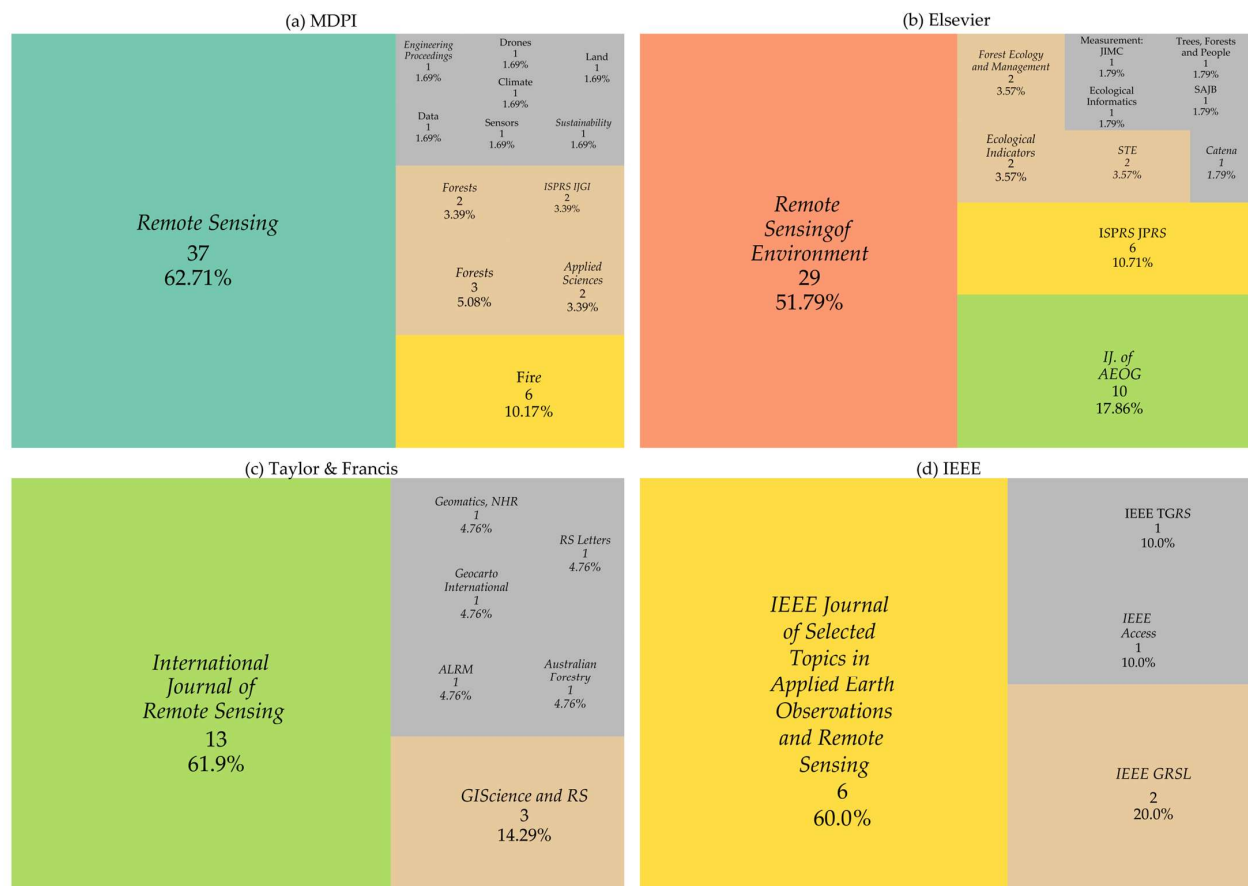


Figure 6. Number of publications by publisher and the percentage each represents in the total studies analyzed: (a) MDPI, (b) Elsevier, (c) Taylor & Francis, and (d) IEEE.

3.1.3. Authors and Citation Numbers

Among the 192 articles reviewed, a total of 754 authors contributed, with several appearing more than once. For instance, Seydi, Seyd Teymoor participated in seven studies, while Fernández-Manso, A. and Quintano, Carmen each published six studies. The names of authors who have significantly contributed to research on fire mapping, highlighting those with five or more publications, are depicted in Figure 7. The complete list of authors is presented in Supplementary Material Table S1.

Regarding the most influential publications based on citation numbers (reviewed as of 30 December 2024), the following stand out: “Global estimation of burned area using MODIS active fire observations” [48], with 484 citations; “Comparison of burn severity assessments using differenced normalized burn ratio and ground data” [49], with 374 citations; and “SENTINEL-2A red-edge spectral indices suitability for discriminating burn severity” [50], with 303 citations. The details of the 10 most-cited studies are presented in Table 2.

Table 2. The 10 most-cited documents on fires worldwide based on Scopus from 1990 to 2024.

Authors	Year	Title	Journal	Number of Citations
Giglio et al. [48]	2006	“Global estimation of burned areas using MODIS active fire observations”	<i>Atmospheric Chemistry and Physics</i>	484
Cocke et al. [49]	2005	“Comparison of burn severity assessments using Differenced Normalized Burn Ratio and ground data”	<i>International Journal of Wildland Fire</i>	374

3.2. Approaches and Tools for Fire Mapping

3.2.1. Sensors

Platforms equipped with active sensors based on SAR and LiDAR, visualized in Figure 8a, have significantly contributed to fire mapping. Notable among these are sensors and missions such as TerraSAR-X, NISAR, RADARSAT, ALOS PALSAR, POLSAR systems, and the Velodyne VLP-16 LiDAR sensor. Conversely, the use of passive sensors has been more prevalent due to their wide availability over the years. Commonly utilized multispectral sensors include MSI, OLI, TM, ETM+, MODIS, AVHRR, TIRS, SPOT, and ASTER and the commercial sensors Dove Classic, Dove-R, and SuperDove from the Planet constellation. In the case of hyperspectral sensors, notable examples include G-LiHT, APEX, PRISMA, Hyperspectral Probe I, and HySpIRI, all of which are depicted in Figure 8b. Additionally, thermal and infrared sensors, which are essential for the active detection of hotspots and surface temperature estimations, include MODIS, ASTER, TIRS, AVHRR, SEVIRI, ABI, and HySpIRI, among others, and are described in Figure 8c.

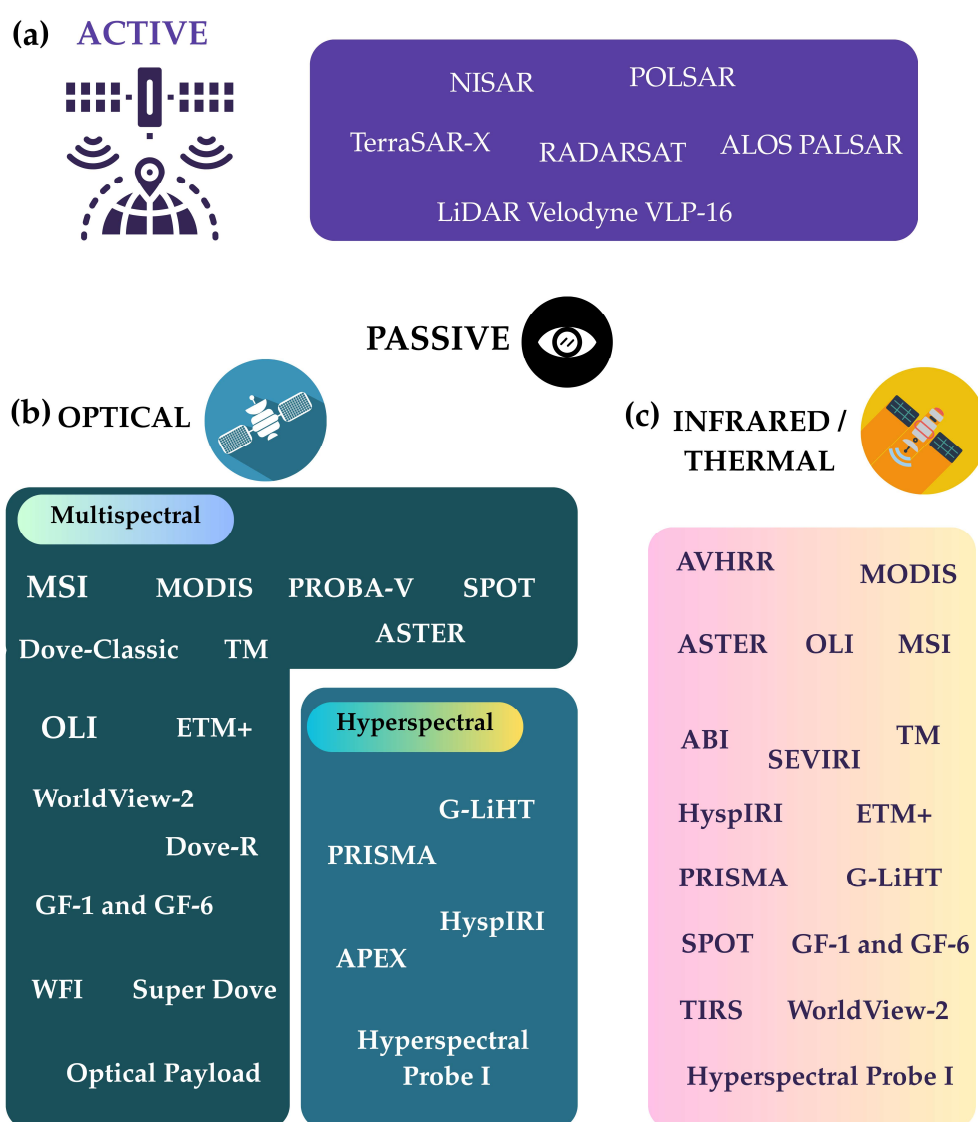


Figure 8. General classification of sensors used in wildfire mapping studies: (a) active sensors; (b) optical multi- and hyperspectral sensors; (c) thermal and infrared sensors.

Furthermore, the most commonly utilized satellites, either individually or through data combination, are presented in Figure 9. Noteworthy among these are the Landsat con-

stellation, which was employed in 55 studies; Sentinel in 43 studies; and MODIS onboard the Aqua and Terra satellites in 10 studies. Less frequently used were PlanetScope images (four studies), Suomi NPP (two studies), and other satellites such as PRISMA, WorldView, NOAA-11, Himawari-8, SPOT-4, and CBERS-4, each appearing in one study. It is important to note that 32 studies utilized satellite images obtained from a combination of multiple satellites. However, this approach is not very common, as each specific combination of satellites was only reported once in the reviewed literature.

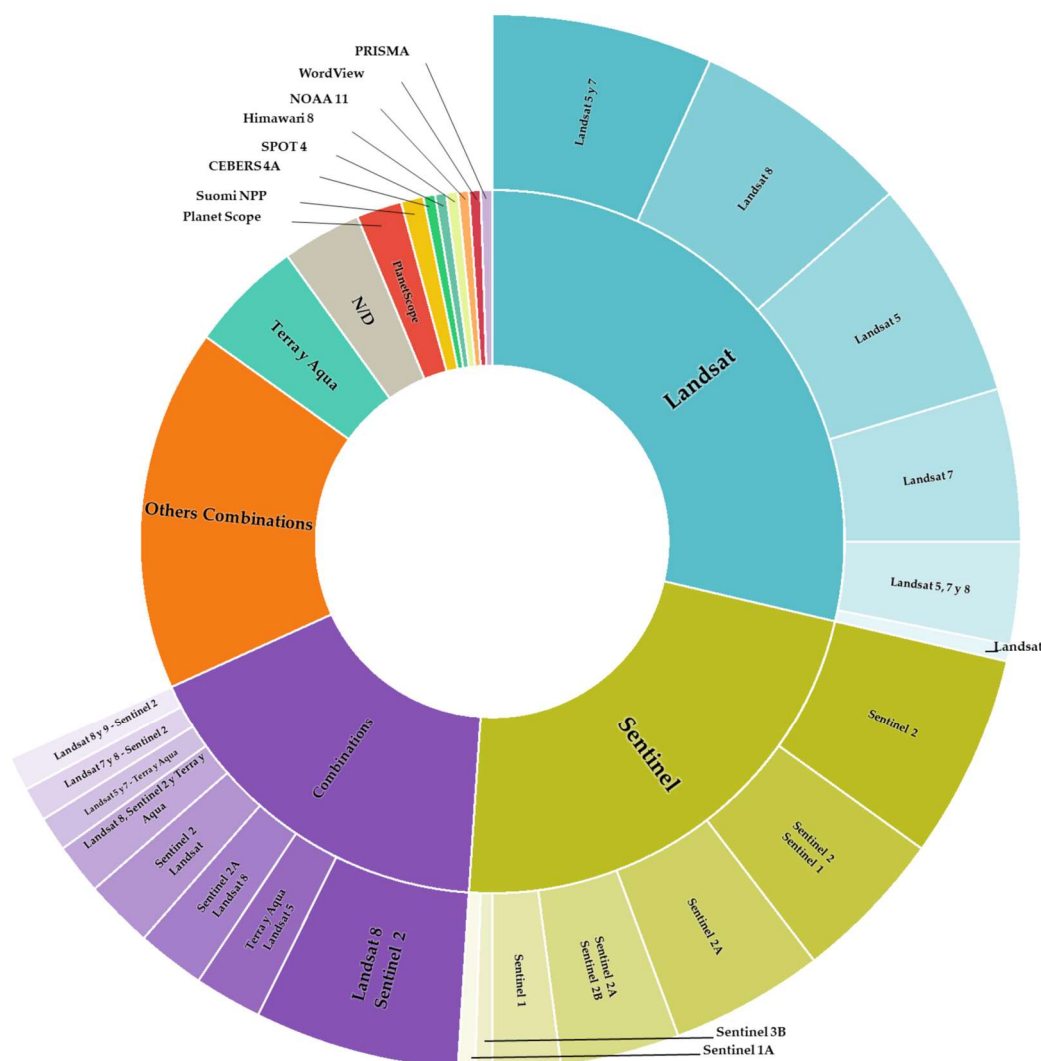


Figure 9. Satellites most utilized in fire mapping from 1997 to 2024.

On the other hand, the use of drones has increased in recent years. In this review, nine studies employed UAVs not only to map burned areas using maximum likelihood and segmentation approaches but also to assess fire severity and monitor post-fire vegetation regrowth. Most of these studies relied on multirotor platforms such as DJI Phantom 4 Pro, Mavic 2 Enterprise Dual, and hexacopters or octocopters due to their stability in post-fire environments. Regarding sensors, RGB cameras were frequently used for basic mapping and classification, while multispectral cameras (used in eight studies) allowed more detailed quantification of burn severity. In one study, multispectral data were integrated with LiDAR to obtain 3D structural information. The use of UAVs facilitated accurate burned area detection, fire severity assessment, and high-resolution monitoring of post-fire vegetation recovery.

3.2.2. Spectral Indices

The NBR was the most utilized in fire mapping and was present in 81 articles (42.19%) of the 192 evaluated, followed by the NDVI with 75 studies (39.06%), and the composite burn index (CBI) with 35 articles (18.23%). Table 3 and Table S2 presents the main indices employed, including the set of other indices that collectively represent 29.17% of the total.

Table 3. Spectral indices used for fire mapping, with the percentage value representing the total number of articles.

Acronym (%)	Index Name	Formula	Reference
NBR (42.19)	Normalized Burn Ratio	$\frac{NIR - SWIR}{NIR + SWIR}$	Key & Benson, 1999 [58]
NDVI (39.06)	Normalized Difference Vegetation Index	$\frac{NIR - R}{NIR + R}$	Rouse et al., 1974 [59]
CBI (18.23)	Composite Burn Index	$NDBI - NDVI - MNDWI$	Key & Benson, 2006 [60]
MIRBI (10.42)	Mid-Infrared Burn Index	$10 LSWIR - 9.8 SSWIR + 2$	Trigg & Flasse, 2001 [61]
BAI (8.85)	Burned Area Index	$\frac{NIR - R}{NIR + R + 0.16}$	Huete, 1988 [62]
SAVI (8.85)	Soil-Adjusted Vegetation Index	$\frac{1}{(0.1 + R)^2 + (0.06 + NIR)}$	Chuvieco et al., 2002 [63]
EVI (7.29)	Enhanced Vegetation Index	$2.5 \frac{(NIR - R)}{NIR + 6R - 7.5B - 1}$	Huete et al., 2002 [64]
NBR2 (5.73)	Normalized Burn Ratio 2	$\frac{SWIR1 - SWIR2}{SWIR1 + SWIR2}$	Key & Benson, 2006 [60]
NDMI (4.69)	Normalized Difference Moisture Index	$\frac{NIR - sSWIR}{NIR + sSWIR}$	Wilson & Sader, 2002 [65]
MSAVI (4.17)	Modified Soil-Adjusted Vegetation Index	$\frac{2(NIR + 1) - ((2NIR + 1)^2 - 8(NIR - R)^{0.5})}{2}$	Qi et al., 1994 [66]
GEMI (4.17)	Global Environment Monitoring Index	$\gamma(1 - 0.25\gamma) - \frac{R - 0.125}{1 - R}$	Pinty & Verstraete, 1992 [67]
Geo CBI (3.13)	Geospatial Composite Burn Index	$\frac{\sum_{m=1}^n (CBI_m \times FCOV_m)}{\sum_{m=1}^n (FCOV_m)}$	De Santis & Chuvieco, 2009 [68]
NDWI (2.60)	Normalized Difference Water Index	$\frac{NIR - G}{NIR + G}$	Gitelson et al., 1996 [69]
GNDVI (2.60)	Green Normalized Difference Vegetation Index	$\frac{Green - NIR}{Green - NIR}$	McFeeters, 1996 [70]
BAIS2 (2.08)	Burned Area Index for Sentinel-2	$\left(1 \times \sqrt{\frac{B06 \times B07 \times B8A}{B4}}\right) \times \left(\frac{B12 - B8A}{\sqrt{B12 + B8A}} + 1\right)$	Filipponi, 2018 [71]
Others (29.17)	Other indices		

3.2.3. Software and Processing

The analysis of software usage and platforms for remote sensor data processing is presented in Figure 10. Among the most frequently used software, ENVI, ArcGIS, and QGIS were mentioned in 52 studies. Additionally, cloud platforms such as GEE, Google Colaboratory, and LANDFIRE were employed in 39 articles. Although various authors integrated tools from different sources by combining programs and platforms complementarily, 101 studies did not specify the use of software or geospatial analysis platforms.

Among the approaches employed for wildfire mapping through artificial intelligence, machine learning has been established as a robust technique for analyzing large volumes of geospatial data, mentioned in 13 studies. The most utilized algorithms were ordinal logistic regression (seven studies), linear regression (six), simple regression (five), and regression tree models (four). Additionally, the use of algorithms such as extreme learning machine, SVM, geographically weighted regression (GWR), Bayesian ZOIB models, decision trees

with spatial autocorrelation, gradient boosting regression, and generalized linear models (GLM) was each reported in one study.

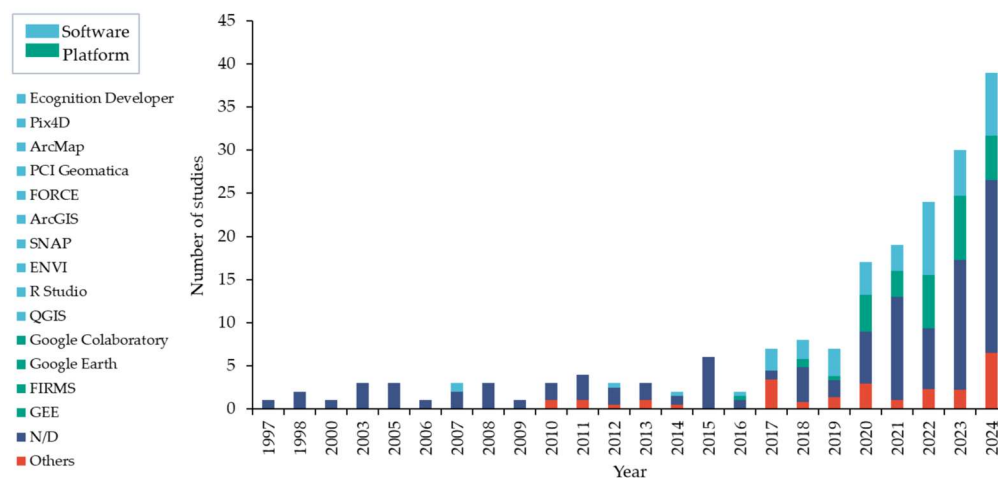


Figure 10. Use of software and platforms used for fire studies between 1997 and 2024.

Regarding deep learning, its use was highlighted as a key tool for identifying complex patterns with high precision, being generally mentioned in 10 studies. Deep neural networks were utilized in two studies, convolutional neural networks in one, and specialized models such as the deep Siamese morphological network in one study. It is noteworthy that a total of 124 studies employed more than one machine learning or deep learning algorithm or a combination of both, with the aim of comparing their performance or complementing analyses conducted for wildfire mapping.

4. Discussion

The incorporation of new remote sensing technologies has significantly transformed environmental monitoring, with wildfire mapping being one of the most beneficial applications. The increase in scientific publications related to this topic since 2015 coincides with global trends previously reported in bibliometric analyses focused on remote sensing and wildfires [72] as well as studies on urban biodiversity [73]. This growing interest in the applications of remote sensing for forest fire monitoring growth can be attributed to several factors. Notably, the launch of Sentinel-2A (2015) and Sentinel-2B (2017) satellites, followed by Landsat 9 in 2021, marked significant technological breakthroughs that enhance the spatial, spectral, and temporal resolution of Earth observation data. With revisit times of about 5 days for Sentinel-2 and 8 days for the combined Landsat 8–9, these missions have improved access to high-resolution imagery for fire monitoring [74]. Additionally, PlanetScope data, available globally since 2016 with near-daily coverage [75], has expanded the ability to detect wildfires, map burned areas, and assess post-fire impacts. Together, these advances have enabled more frequent and accurate analyses of fire dynamics, likely contributing to the rise in scientific publications after 2015, as shown in Figure 4. In addition, the development of platforms such as GEE have played a key role in democratizing access to large volumes of historical and current data, enabling integrated analyses with machine learning algorithms, which has favored the early and accurate detection of wildfires, as mentioned by Pérez-Cutillas et al. [76].

The number of scientific publications on forest fire mapping varies considerably between countries due to the influence of economic, social, and technological factors in each country. Countries such as the United States and Brazil stand out for their substantial investments in space infrastructure, customized satellite programs, and strong international collaboration. The United States, one of the pioneers in this field, has been at the forefront

since the launch of the first artificial satellite and the creation of the National Aeronautics and Space Administration (NASA). Through NASA, this country has access to satellites such as Landsat and MO-DIS, which have been fundamental for monitoring forest fires. Similarly, China has developed satellites such as Gaofen (launched in April 2013) and Jilin-1 (launched in October 2015), improving the capacity for Earth observation and monitoring. For its part, Brazil increased its number of publications on forest fires due to the work of the Instituto Nacional de Pesquisas Espaciais (INPE), which manages satellites such as CBERS and the DETER platforms, which monitor deforestation and fires in the Amazon in real time, allowing easy access to researchers.

However, countries with a small number of scientific publications need to improve their learning systems and incorporate new techniques to address this problem. This is reflected in the studies of Von Essen et al. (2025) [77], who, through a survey of public officials, pointed out that unlike Brazil, neighboring countries such as Peru face serious limitations in terms of resources and experience in these issues. In addition to the lack of advanced techniques, it is often necessary to carry out field observations and analysis, which presents great challenges, especially in regions such as the African tropics, where conditions make it difficult to implement effective monitoring technologies. Furthermore, the low level of scientific publications is also observed in countries with minimal fire incidence, such as Sweden. This country, with a low incidence of forest fires during the 20th and early 21st centuries [78], has shown limited interest in research on this topic, which is reflected in the low scientific production in the region. It should be noted that the neglect of the environment coupled with the lack of investment in technology and training to address the factors that cause forest fires significantly limits the production of scientific research. This lack of resources hinders an adequate response to fires and increases inequality in the number of studies, especially in countries with less infrastructure. As a result, some countries publish more scientific articles on forest fire mapping due to their access to advanced technologies and consolidated research systems, while others with fewer resources face difficulties that prevent the development of effective strategies for the prevention and monitoring of these events.

From a publishing standpoint, MDPI is the leading publisher with the highest volume of articles on the subject, particularly in journals such as *Remote Sensing and Fire*, which offer short acceptance times (17 to 24 days). In contrast, publications such as *Remote Sensing of Environment* (Elsevier) and *International Journal of Applied Earth Observation and Geoinformation* have more extended review processes (138 to 203 days). Publishers like Taylor & Francis and IEEE publish articles exclusively focused on wildfires less frequently, which may be related to their more technical orientation [79]. Regarding authors, S.T. Seydi stands out for their work with deep learning and Sentinel-2 data, while C. Quintano and A. Fernández-Manzo are noted for their contributions to regional mapping using hyperspectral images and forest inventories. Researchers such as Y. Ban, L. Boschetti, E. Chuvieco, and others have contributed to the use of advanced algorithms and emerging sensors, whereas Chanussot and Fernández Guisuraga have innovated in the use of UAVs and deep neural networks. The thematic and methodological diversity explains the variability in the number of publications among authors.

The use of Landsat satellites is remarkably frequent due to their long observation history, which began in 1972 with the ERTS-1 (Landsat 1) satellite [80], providing continuous data until today, with its latest launch of the satellite Landsat 9 (2021), which will be complemented by Landsat Next, expected to launch in late 2030/early 2031 (27 September 2021). In addition, Landsat offers images with a spatial resolution of 30 m, which considerably improves accuracy in identifying areas affected by fires [81]. Likewise, multiple studies highlight positive aspects regarding the use of this satellite at various time

scales [82,83] as well as the fact that this satellite is an open-access scientific source [84], making it possible to have a greater number of scientific publications that use this satellite, in accordance with the data shown in Figure 10.

The European Space Agency (ESA) Sentinel-1 and Sentinel-2 satellites have become crucial tools for wildfire detection, each with characteristics that make it suitable for different conditions [85]. Sentinel-1, which uses radar, is mostly used in tropical areas because of its ability to operate independently of meteorological conditions, such as cloud cover, making it ideal for regions where optical visibility is limited [85]. In addition, it helps to evaluate changes in vegetation structure; however, these data are influenced by the physical properties of the soil, which requires field validation to ensure the accuracy of the results [86,87]. Therefore, the effectiveness of Sentinel-2, which is based on optical sensors, is significantly reduced under cloudy conditions, such as those found in tropical rainforests [88,89]. However, it has shown good performance in wildfire monitoring when integrated with Landsat data [90]. Furthermore, in fire-prone arid and semi-arid regions, optical data are highly effective due to low atmospheric interference and high spectral contrast between burned and unburned areas [91,92].

Likewise, Sentinel-2, with its 10 m spatial resolution and high revisit frequency of 5 days, has demonstrated increased use for accurate detection of forest fires [93]. This capability allows it to provide constant and detailed surveillance, providing optical data that, in combination with those of Sentinel-1, improve accuracy in identifying affected areas. The combination of both satellites with their complementary capabilities has been fundamental to improve coverage and accuracy in forest fire monitoring, especially in tropical areas where the combination of optical and radar technologies is essential to overcome the limitations of each individual system [94,95].

Thus, while Landsat has been a fundamental tool for decades, the addition of the Sentinel-1 and Sentinel-2 satellites significantly expands monitoring capabilities, offering a more robust solution adapted to diverse geographic and climatic conditions. The union of these satellites would significantly improve the accuracy and reliability of damage severity maps, offering a more robust solution than the isolated use of any single satellite.

On the other hand, MODIS, despite providing satellite data with a low resolution (250 m to 1 km), is recognized and widely used due to its high revisit rate, between 1 and 2 days, allowing continuous monitoring in areas affected by fires [96,97]. Also, more recent commercial satellites, such as PlanetScope, Suomi NPP, SEBER 4A, SPOT, and Himawari, offer advantages in terms of revisit frequency. For example, PlanetScope has the ability to reach resolutions of up to 3 m, which allows for more accurate data [98]. However, despite these advantages, the use of these satellites presents challenges such as high costs and the lack of thermal functionalities in some models such as SPOT, which limits their application [99].

Passive optical sensors (MSI, TM, OLI, and ETM+) predominate in fire studies due to their high resolution, which is useful for the delineation of affected areas and post-fire monitoring. MODIS, despite its low resolution, is key for the rapid detection of active fire hotspots [100–102]. Active sensors such as SAR, although less common, effectively complement optical data, especially under cloudy conditions [103]. High-resolution commercial sensors, such as those from Planet Labs, are valuable but less widely used on a large scale due to economic and availability constraints [104]. In recent years, the use of UAVs in the detection of burned areas and post-fire monitoring has increased, providing high-resolution information on affected areas that can be scaled to the pixel footprints of medium-resolution satellite images such as Sentinel or Landsat [105]. UAVs also allow rapid data acquisition immediately after fire events and, in some cases, can even support anticipatory monitoring of fire occurrence [106,107]. Most studies relied on multirotor

UAVs (8/9), primarily equipped with multispectral sensors, whereas LiDAR was used only once as a complementary technology. However, the cost of RGB, multispectral, or LiDAR cameras or sensors may limit their use in addition to the acquisition of images before a wildfire to assess the severity of fires [108].

Among spectral indices, the NBR is the most employed due to its ability to clearly differentiate burned areas, surpassing the NDVI index in fire sensitivity [109]. Other indices, such as CBI and MIRBI, provide information on fire severity [110,111], while SAVI, BAI, EVI, NBR2, and NDMI contribute to the assessment of post-fire condition and residual terrain moisture [112–115].

The application of specialized software has significantly enhanced analytical capabilities. Tools such as QGIS with the Semi-Automatic Classification Plugin—SCP plugin [116], ENVI, ArcGIS, R Studio (various versions), and SNAP facilitate the integration, analysis, and modeling of satellite data [117,118]. GEE provides a robust platform for cloud-based geospatial analysis, while FIRMS specializes in real-time wildfire monitoring [119]. On the other hand, Google Colab streamlines the implementation of machine learning models in the cloud using python language. There are four main approaches for calculating biophysical parameters in burned areas: parametric and non-parametric regression models, empirical models, and physics-based models. In the context of machine learning, algorithms such as RF, SVM, and non-parametric neural networks have shown great efficacy in capturing non-linear relationships and handling complex datasets [120–124]. DL models, such as CNN, are notable for their ability to estimate fire severity and vegetation recovery, especially when combined with other algorithms like XGBoost [125–128].

Despite technological advancements and the increasing availability of Earth observations data, several limitations remain. A major constraint is the high cost of commercial high-resolution satellite imagery, which limits its accessibility for research institutions and operational agencies in low-resource settings. Additionally, the processing of data from active sensors such as SAR or LiDAR, which are necessary to overcome the limitations of passive sensors in areas with high cloud cover, requires substantial computational resources and storage infrastructure. In this context, artificial intelligence offers promising solutions for automating image processing and integrating data from various platforms and sensors, enhancing the monitoring and mapping of areas affected by fires and other extreme events. Emerging models such as 3D convolutional neural networks (e.g., U-Net 3D) and transformer-based architectures have shown potential for improving accuracy in burned area detection and real-time fire prediction. Future research should also explore cloud-based data fusion as well as blockchain-enabled platforms for transparent tracking of firefighting resources and post-fire recovery actions. Moreover, the integration of drones can enhance fire monitoring in areas that are difficult to access, contributing to more efficient and resilient wildfire management strategies.

5. Conclusions

This review highlights the impact of remote sensing technologies such as Landsat, Sentinel-1, and Sentinel-2 satellites on forest fire monitoring, improving accuracy and coverage since 2015. However, there are still challenges, especially in countries with fewer resources, such as Peru and Sweden, which make it difficult to implement effective methods. Technologies such as LiDAR and orthophotos play a crucial role in post-fire analysis, with LiDAR providing detailed 3D data and orthophotos enabling accurate visualizations. However, their use is limited by costs and the need for flights.

In the environmental field, it is essential to take into account the specific characteristics of the ecosystem when selecting the most appropriate technologies. In tropical ecosystems, where weather conditions complicate the use of optical satellites, radar sensors such as

Sentinel-1 are more effective. For more detailed vegetation analysis, a combination of Landsat and LiDAR can provide a more complete view of the terrain and post-fire effects. In addition, open-access platforms such as Google Earth Engine are essential for managing large volumes of historical and current data without restrictions, which improves the efficiency of the analysis.

Early fire prediction using advanced models and machine learning can help identify high-risk areas and forecast the spread of fire, facilitating early decision making and reducing its impact. Neural networks and other machine learning techniques optimize the detection of affected areas, especially in complex terrain. Fusion of satellite data with cloud-based geoprocessing platforms improves the accuracy and speed of predictions. In addition, technologies such as LiDAR and SAR radar make it possible to map affected areas in difficult conditions such as cloud cover or complex terrain.

This review provides researchers with an overview of current technologies, facilitating the selection of appropriate methodologies according to local conditions. The integration of advanced modeling, machine learning, drones, and cloud analytics improves monitoring, early detection, and real-time response and as such is essential for post-fire management and prediction of future events, which helps to protect ecosystems and reduce risks. In future research, it would be beneficial to study the use of blockchain in disaster management, improving efficiency and transparency in the allocation of resources and in the transmission of information between entities responsible for fire management. Likewise, the use of the Radar Observation System for Europe L-Band (ROSE-L), developed by the European Space Agency (ESA), has great potential to contribute to forest fire monitoring. This synthetic aperture radar (SAR) system is capable of assessing geological hazards; tracking changes in land use, agriculture, and forestry; and collecting accurate data on soil moisture. Its application in forest fires could be essential to detect fires in their early stages, perform damage analysis, and support decision making to improve response and management of available resources.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/fire8080316/s1>. Table S1: List of the 192 selected articles used in the review; Table S2: Spectral indices used for fire mapping; the percentage value is with respect to the total number of items.

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